COMSATS University Islamabad, Attock Campus

Department of Computer Science

Course: Artificial Intelligence

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Lab Task	

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Question 1:

Write a Python function to calculate GCD using Euclid's Algorithm.

Code:

```
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a

print("GCD:", gcd(48, 18))
```

Output:

```
PS D:\python code\Information Secrity
/Information Secrity Lab/Lab 11/Task1.
GCD: 6
PS D:\python code\Information Secrity
```

Question 2:

GCD with Step DisplayModify the GCD function to display each step of Euclid's Algorithm. (5 Marks)

Code:

```
def gcd_with_steps(a, b):
    print("Steps of Euclid's Algorithm:")
    while b != 0:
        print(f"{a} = {b} × ({a // b}) + {a % b}")
        a, b = b, a % b
    print(f"GCD = {a}")
    return a
```

```
gcd_with_steps(48, 18)
```

Output:

```
Steps of Euclid's Algorithm:
48 = 18 × (2) + 12
18 = 12 × (1) + 6
12 = 6 × (2) + 0
GCD = 6
PS D:\python code\Information Secrity Lab\Lab 11>
```

Question 3:

Recursive Implementation (5 Marks) Implement Euclid's Algorithm using recursion instead of a loop.

Code:

```
def gcd_recursive(a, b):
    if b == 0:
        return a
    return gcd_recursive(b, a % b)


print("GCD (Recursive):", gcd_recursive(48, 18))
```

Output:

```
PS D:\python code\Information Secri
/Information Secrity Lab/Lab 11/Tas
GCD (Recursive): 6
PS D:\python code\Information Secri
```